

LLM Capability Is Not a Scalar: Measuring the Shape of Reasoning-Model Behavior with Path Consistency and Sampling Fingerprints

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Abstract

Reasoning models are compared almost entirely by a single accuracy number, typically Pass@1 on a benchmark. But this number is a lossy projection: it collapses how a model produces an answer down to whether its first attempt is correct, discarding information that bears directly on deployment. We show that this lost structure is real by holding accuracy fixed and asking what still varies: two models with nearly identical Pass@1 can differ sharply in how *consistently* they reason across repeated attempts, and in how their accuracy *scales* as attempts accumulate. We introduce two complementary instruments for measuring it. **Reasoning Path Consistency** (RPC) quantifies how similar a model’s reasoning paths are across repeated attempts, along lexical, structural, and semantic axes, with its semantic variant validated against human judgments of reasoning similarity. **Parametric Sampling Fingerprints** (PSF) fit a three-parameter curve to a model’s Pass@ k trajectory, summarizing how its accuracy grows with additional samples. Across sixteen models and four reasoning benchmarks, models with matched Pass@1 diverge substantially under both instruments—semantic path consistency differs by up to 0.17, and Pass@4 by up to 10 points—and the two instruments are statistically independent, capturing distinct dimensions of behavior. Capability, we argue, is not a scalar: it has a shape, and measuring that shape reveals differences that accuracy alone conceals.

1 Introduction

Progress in reasoning models is overwhelmingly reported as a single number: accuracy, or Pass@1, on a benchmark (Li et al., 2026b; Chen et al., 2021; Paul et al., 2024; Pathak et al., 2026). This number is convenient and comparable, but it is also lossy. It says nothing about *how* a model arrives at its answer, whether it would arrive at the same answer if asked again, or how its performance would change

if it were allowed more attempts (Yang et al., 2026; Massoli et al., 2026). Two models can sit at the same point on a leaderboard and behave entirely differently underneath (Zhao et al., 2026; Wang et al., 2025).

We make this concrete. On a law-reasoning benchmark, Claude-4.5-Sonnet and Gemini-2.5-Pro achieve first-attempt accuracies that differ by less than half a percentage point—they are, by the standard metric, the same. Yet the consistency of their reasoning paths across repeated attempts differs by 0.17 on a 0-1 scale, a gap that is overwhelmingly statistically significant. On a mathematics benchmark, GPT-OSS-120B and GPT-5-mini are likewise tied on Pass@1, but by the fourth attempt their accuracies differ by more than ten points, because one model saturates almost immediately while the other keeps improving with more samples. In both cases, the scalar accuracy is identical and the behavior is not.

We argue that *capability is not a scalar*: a model’s behavior on a benchmark has a shape that a single accuracy number cannot express, and that shape is measurable. We introduce two complementary instruments for measuring it, and Figure 1 summarizes how they relate.

Reasoning Path Consistency (RPC). Given a model’s reasoning traces across repeated attempts at the same problem, RPC measures how similar those paths are, along three axes: lexical overlap (RPC-L), structural similarity of step types (RPC-S), and semantic similarity of step content (RPC-SEM) (Wu et al., 2026). The three axes are correlated but distinct, and RPC-SEM, our primary axis, is validated against human judgments of whether two reasoning paths take “the same approach” (Spearman $\rho = 0.34$, $n = 50$, $p = 0.017$; protocol in Appendix C).

Parametric Sampling Fingerprints (PSF). Given a model’s empirical Pass@ k for

$k = 1 \dots 4$, PSF fits the three-parameter curve $\text{Pass}@k = \alpha + (\beta - \alpha)(1 - e^{-\gamma(k-1)})$, where α is first-attempt accuracy, β is the saturation ceiling, and γ governs how quickly the model approaches it. The fit is excellent (mean $R^2 = 0.96$ over non-degenerate fits), making the three parameters a compact fingerprint (Pasquini et al., 2025) of a model’s sampling behavior.

Findings. Across sixteen models and four benchmarks (law, mathematics, graduate-level science, and multi-modal reasoning), we find: **(1)** models with matched Pass@1 diverge substantially in reasoning-path consistency, with semantic gaps up to 0.17 at near-identical accuracy; **(2)** models with matched Pass@1 diverge substantially in sampling behavior, with Pass@4 gaps up to 10 points; **(3)** RPC and PSF are statistically independent, confirming that reasoning consistency and sampling trajectory are distinct dimensions of behavior rather than two views of the same signal; and **(4)** both forms of structure are domain-conditional—a model’s behavior shape changes across reasoning domains, most strongly in the structural organization of its reasoning.

Contributions.

- We introduce **RPC**, a three-axis measure of reasoning-path consistency across repeated attempts, with its semantic axis validated against human judgment.
- We introduce **PSF**, a three-parameter fingerprint of a model’s Pass@ k scaling behavior that fits empirical sampling curves with high fidelity.
- We show that both instruments reveal large behavioral differences between models that Pass@1 reports as equivalent, and that the two instruments are independent, together providing a more complete picture of model behavior than accuracy alone.

2 Related Work

Beyond single-number evaluation. A growing body of work argues that benchmark accuracy alone is an inadequate summary of model capability. Holistic evaluation efforts broaden the set of reported metrics across many scenarios (Bommasani et al., 2023; Li et al., 2026a), and analyses

of benchmark contamination, saturation, and construct validity have questioned what leaderboard accuracy actually measures (Bean et al., 2025; Akhtar et al., 2026; Sander et al., 2025). This work shares that motivation but takes a different cut: rather than adding more scenarios or more accuracy-like metrics, we characterize the *behavioral shape* behind a fixed accuracy—how consistently a model reasons and how its accuracy scales with sampling—and show that this shape varies sharply among models that accuracy reports as equivalent.

Self-consistency and repeated sampling. Repeated sampling is central to modern inference. Self-consistency improves reasoning accuracy by sampling multiple chains and marginalizing over reasoning paths to a majority answer (Zhou et al., 2025; Feng et al., 2025), and broader analyses of repeated sampling show that Pass@ k can scale substantially with the number of samples (Qi et al., 2025; Brown et al., 2025). These methods treat the multiple samples as a means to a better *answer*; we instead treat them as a window into model *behavior*. RPC measures whether the sampled reasoning paths are mutually consistent, and PSF measures the rate at which additional samples convert into additional correct answers—two properties of the sampling process itself, orthogonal to its use as an accuracy-boosting technique.

Pass@ k and scaling behavior. Pass@ k is widely reported, particularly in code generation (Tang et al., 2024; Chen et al., 2021; Kullal et al., 2019; Chen et al., 2026), and recent work characterizes how performance scales with inference-time compute and repeated attempts (Chen et al., 2025; Brown et al., 2025; Thrampoulidis et al., 2026; Levi, 2024). These analyses typically report Pass@ k at a few fixed k or study aggregate scaling trends. PSF instead fits a compact parametric form to each model-benchmark Pass@ k curve, yielding interpretable parameters—first-attempt accuracy, saturation ceiling, and approach rate—that serve as a per-model fingerprint and make two models’ sampling behavior directly comparable even when their Pass@1 is identical.

Reasoning-path and consistency analysis. Several lines of work examine the internal structure of model reasoning—faithfulness of chain-of-thought to the final answer (Shen et al., 2026; Zhou et al., 2026; Xiong et al., 2025), the effect of reasoning-

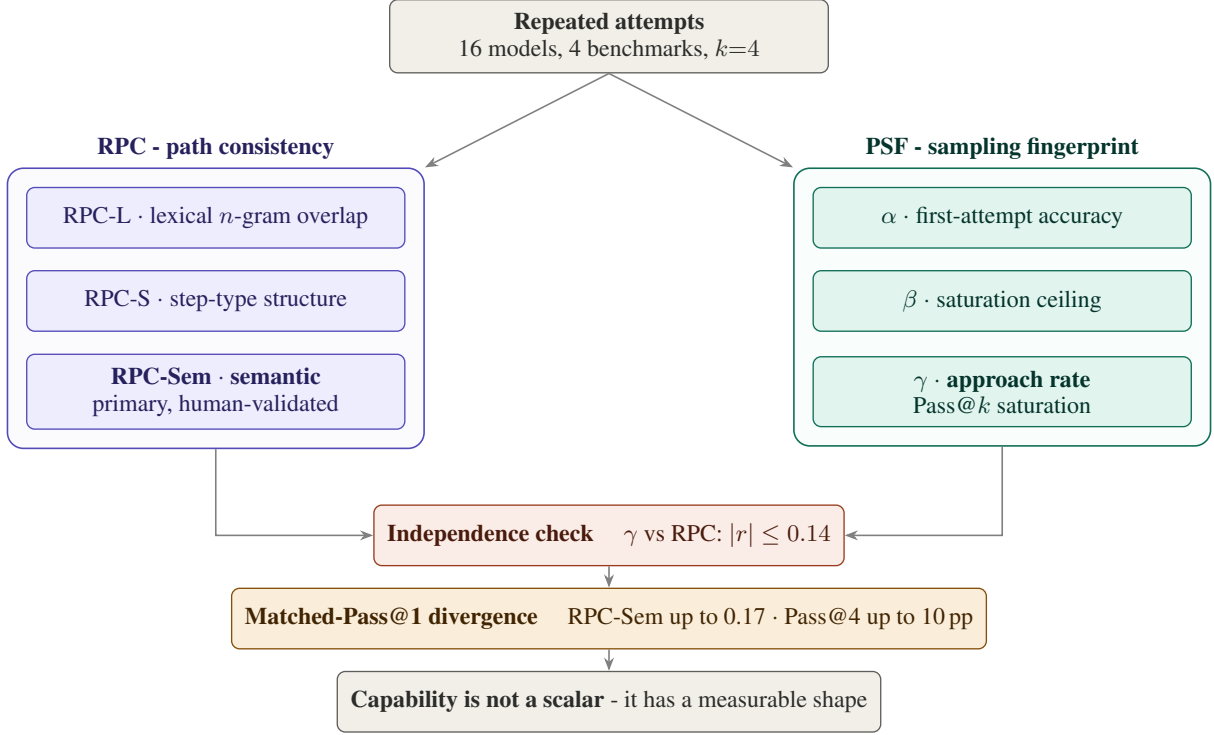


Figure 1: Overview of our approach. A single set of repeated model attempts (sixteen models, four benchmarks, $k = 4$) yields reasoning traces, which feed **RPC** (lexical, structural, and semantic path consistency), and correctness labels, which feed **PSF** (the α, β, γ sampling fingerprint). The two instruments are statistically independent and both reveal behavioral divergence between models matched on Pass@1, supporting the thesis that capability is not a scalar.

chain perturbations, and stability of outputs under paraphrase or resampling. Work on output consistency typically asks whether a model produces the same *answer* under perturbation. RPC asks a finer question: whether the model traverses the same *path* to its answers across attempts, decomposed into lexical, structural, and semantic similarity. To our knowledge, reasoning-path consistency across repeated attempts has not previously been formalized as a multi-axis, human-validated metric and used to distinguish models at matched accuracy.

Model fingerprinting and behavioral comparison. Prior work has sought to identify or distinguish models from their outputs-stylometric and watermark-based fingerprinting, and behavioral comparison of model families (Pasquini et al., 2025). RPC and PSF can be read as behavioral fingerprints, but our goal is not identification: it is measurement of behavioral *dimensions*-reasoning consistency and sampling trajectory-that are interpretable, comparable across models, and, as we show, statistically independent of one another and of first-attempt accuracy.

Summary. Existing work largely either enriches *what* is evaluated by adding scenarios and met-

rics (Bean et al., 2025; Ahn et al., 2026), uses repeated sampling to improve *answers* (Tang et al., 2024; Wei et al., 2022), or studies aggregate scaling trends. We instead introduce two interpretable, independent instruments-one for reasoning-path consistency, one for Pass@ k scaling shape-and use them to show that models tied on Pass@1 differ systematically in how they reason and how they scale.

3 Experimental Setup

Both instruments operate on the same underlying data-a set of repeated model attempts at each problem-which we describe here. For each model and benchmark, every problem is attempted $k = 4$ times under independent sampling (Wang et al., 2023), yielding four full responses per problem. Each response contains the model’s reasoning trace and its final answer, and is graded against the benchmark’s gold answer to produce a per-attempt correctness label. RPC operates on the four reasoning traces; PSF operates on the four correctness labels.

Models and benchmarks. We evaluate sixteen models spanning seven developers (Anthropic, Google, OpenAI, DeepSeek, xAI, MiniMax, and Qwen) across four reasoning benchmarks: law

(AGIEval LSAT-AR) (Zhong et al., 2023), mathematics (AIME 2025) (Mathematical Association of America, 2025), graduate-level science (GPQA Diamond) (Rein et al., 2023), and multimodal reasoning (MMMU subsets) (Yue et al., 2024). Benchmarks, the full model roster, and per-benchmark coverage are detailed in Appendix B (Tables 7 and 8).

Reasoning-trace availability. RPC requires access to a model’s reasoning trace, not only its final answer (Zhang et al., 2026; Twist et al., 2026). Models in the OpenAI family (o4-mini, GPT-5, GPT-5-mini, GPT-5-nano, GPT-5.1, and GPT-OSS-120B) do not expose intermediate chain-of-thought through their API, returning only final answers. We therefore *exclude these models from all RPC analyses*, which is a principled scoping decision rather than a data-quality artifact: RPC is undefined without reasoning traces. PSF, which requires only correctness labels, retains all sixteen models. This asymmetry is intrinsic to what each instrument measures and we report each instrument over the set of models for which it is defined (Table 7).

4 Metrics: RPC and PSF

We introduce two instruments that operate on the same raw material—a model’s repeated attempts at a problem (Section 3)—but measure different properties of its behavior. **Reasoning Path Consistency** (RPC; Section 4.1) measures how similar a model’s reasoning *paths* are across attempts. **Parametric Sampling Fingerprints** (PSF; Section 4.2) measures how a model’s *accuracy* scales as attempts accumulate.

4.1 Reasoning Path Consistency (RPC)

Given the set of reasoning traces $\{r_1, r_2, r_3, r_4\}$ a model produces on a problem, RPC measures their mutual similarity, averaged over all $\binom{4}{2} = 6$ pairs. We measure similarity along three complementary axes.

RPC-L (lexical). For each pair of traces, we compute the overlap of their token n -grams ($n = 3$) (Papineni et al., 2002; Gusfield, 1997), capturing surface-level reuse of phrasing. RPC-L is reported as the mean pairwise n -gram overlap across the six trace pairs.

RPC-S (structural). Each trace is abstracted to its sequence of step types (e.g. reasoning, tool call, observation), and RPC-S measures the similarity

of these step-type sequences across attempts (Aho and Ullman, 1972; Rasooli and Tetreault, 2015; Reimers and Gurevych, 2019; Chandra et al., 1981). It captures whether the model organizes its solution in the same way across attempts, independent of the specific content.

RPC-SEM (semantic). Each trace is embedded with a sentence-embedding model (all-MiniLM-L6-v2) (Wang et al., 2020), and RPC-SEM is the mean pairwise cosine similarity of the four embeddings. It captures whether the model reasons about the problem in semantically similar terms across attempts, independent of exact wording or step structure. RPC-SEM is our primary axis: it is the most robust to superficial variation and, as we show in Section 5.1, the axis most aligned with human judgments of reasoning similarity.

Validation. Because RPC is a new metric, we validate it against human judgment (Appendix C) (Graham and Baldwin, 2014; American Psychological Association, 1983). Annotators labeled whether two reasoning paths take “the same approach” on a set of 50 trace pairs, and we correlate these judgments with each RPC axis. RPC-SEM correlates with human judgment at Spearman $\rho = 0.34$ ($n = 50$, $p = 0.017$); the lexical and structural axes capture related signal but are weaker proxies for human similarity judgments at this scale, and we therefore treat RPC-SEM as the primary human-aligned axis (Section 5.1). Appendix C gives full computation protocols, the strict-row filter, and supplementary RPC breakdowns.

4.2 Parametric Sampling Fingerprints (PSF)

For a given model and benchmark, let $\text{Pass}@k$ denote the probability that at least one of k independent attempts is correct, estimated empirically for $k = 1, \dots, 4$. PSF summarizes this curve with a three-parameter form:

$$\text{Pass}@k = \alpha + (\beta - \alpha) (1 - e^{-\gamma(k-1)}), \quad (1)$$

which follows the same exponential-saturation form as Amer et al. (2026), fit per model-benchmark pair by least squares (Kazdan et al., 2025; Amer et al., 2026; Andrew and Gao, 2007). Curve fits by benchmark are shown in Appendix D (Figures 5, 6, 7, and 8).

The parameters have direct interpretations: α is first-attempt accuracy ($\text{Pass}@1$); β is the saturation

ceiling approached as k grows; and $\gamma > 0$ governs how quickly the model approaches that ceiling—a large γ indicates near-immediate saturation (most of the benefit comes from the first extra sample), while a small γ indicates accuracy that continues to grow with more attempts. The triple (α, β, γ) is a compact fingerprint of a model’s sampling behavior and renders two models’ scaling directly comparable even when their α is identical.

Fit quality and degenerate cases. Equation 1 fits the empirical curves well, but some model-benchmark pairs are degenerate for curve-fitting: when accuracy barely changes across k (so $\alpha \approx \beta$), when γ is pushed to a boundary by the optimizer, or when a benchmark has too few problems to estimate four Pass@ k points reliably (Kazdan et al., 2025). We report fit quality both over all fits and over a *non-degenerate* subset (those with $|\alpha - \beta| > 0.02$, γ away from the optimizer boundary, and at least 30 problems), and base interpretive claims on the non-degenerate subset (Section 5.3). Appendix D reports the fitting procedure, per-benchmark curve fits (Figures 5, 6, 7, and 8), and the full parameter table for all model-benchmark pairs (Tables 10 and 11).

4.3 Independence of the Two Instruments

RPC and PSF are designed to measure different things—path consistency versus accuracy scaling—and we verify empirically that they are not redundant. Joining the two instruments on the model-benchmark pairs for which both are defined, we find that the PSF approach-rate γ is at most weakly correlated with all three RPC axes (Section 5.4). This independence is what justifies reporting both: each instrument captures a dimension of behavior the other does not, and neither is reducible to first-attempt accuracy (Di et al., 2025; Chang et al., 2023).

5 Results

We organize results around the paper’s thesis. We first validate RPC against human judgment and characterize the relationship among its three axes (Section 5.1). We then present the two central findings—that models matched on Pass@1 diverge under RPC (Section 5.2) and under PSF (Section 5.3)—establish that the two instruments are independent (Section 5.4), and show that behavioral shape is domain-conditional (Section 5.5).

5.1 RPC Validation and Axis Structure

RPC-Sem aligns with human judgment. On 50 trace pairs labeled by humans for whether they take “the same approach” (labeling protocol in Appendix C), RPC-SEM correlates with human judgment at Spearman $\rho = 0.34$ ($p = 0.017$; Table 1), consistent with prior work validating semantic similarity metrics against human annotations (Zhang et al., 2020). The lexical and structural axes capture related signal but are weaker proxies for human similarity judgments at this scale; we therefore treat RPC-SEM as the primary human-aligned axis throughout.

The three axes are correlated but distinct. Table 2 reports the pairwise correlations among the three RPC axes over all model-benchmark cells. The axes rise together—lexical, structural, and semantic consistency are positively correlated—but no pair approaches $\rho = 1$. Vocabulary, organization, and meaning are related but separable signals, not three names for the same quantity, which justifies reporting all three rather than collapsing them (strict-row definition in Appendix C), in line with broader evidence that generation-quality and accuracy metrics can vary independently (Chang et al., 2023).

Axis pair	Spearman ρ	Pearson r
RPC-L vs RPC-SEM	0.50	0.29
RPC-L vs RPC-S	0.36	0.30
RPC-S vs RPC-SEM	0.25	0.24

Table 2: Pairwise correlations among the three RPC axes over $n = 3,998$ strict model-question cells (all three axes defined; all $p < 10^{-50}$). The axes are positively correlated but far from identical.

5.2 Models Matched on Pass@1 Diverge in Reasoning Consistency

Our first central finding is that first-attempt accuracy does not determine reasoning-path consistency. For every benchmark we identify model pairs whose Pass@1 differs by at most 0.02 and compare their per-question RPC-SEM distributions. Among non-OpenAI models (those for which RPC is defined), 12 such matched pairs qualify on the law benchmark (Table 3), and they exhibit substantial RPC-SEM gaps despite near-identical accuracy: 6 of 12 pairs differ by at least 0.05 in mean RPC-SEM, 3 by at least 0.10, and 2 by at least 0.15.

The clearest example is Claude-4.5-Sonnet and Gemini-2.5-Pro: their Pass@1 differs by 0.004—statistically indistinguishable—yet their mean RPC-

Comparison	n	Pearson r	p	Spearman ρ
Human vs RPC-SEM	50	0.34	0.017	0.34
Human vs RPC-S	50	0.12	0.40	0.10
Human vs RPC-L	50	-0.12	0.40	-0.03

Table 1: Correlation between human “same approach” judgments and each RPC axis ($n = 50$ matched trace pairs). Only RPC-SEM reaches conventional significance (Spearman $p = 0.017$).

Model A	Model B	Pass@1 gap	RPC-Sem gap
Claude-4.5-Sonnet	Gemini-2.5-Pro	0.004	0.168
Claude-4-Opus	Gemini-2.5-Pro	0.009	0.155
Claude-4.5-Sonnet	Gemini-3	0.012	0.126
Claude-4.5-Sonnet	MiniMax-M2	0.005	0.099
Claude-4-Opus	MiniMax-M2	0.008	0.086

Table 3: Matched-Pass@1 pairs with the largest RPC-SEM gaps on AGIEval LSAT-AR ($|\Delta\text{Pass@1}| \leq 0.02$; non-OpenAI models only; $n = 12$ qualifying pairs). Among these 12 pairs, 6 differ by ≥ 0.05 , 3 by ≥ 0.10 , and 2 by ≥ 0.15 in mean RPC-SEM; Mann-Whitney tests on per-question RPC-SEM reject equality at $p < 10^{-19}$ for all rows shown.

SEM differs by 0.168, and a Mann-Whitney test on their per-question RPC-SEM distributions rejects equality at $p = 8 \times 10^{-76}$ (Table 3). Gemini reaches the same first-attempt accuracy as Claude while taking markedly less consistent reasoning paths across attempts. Two models that a leaderboard reports as equivalent are, in their reasoning behavior, not equivalent at all (Li et al., 2026b).

5.3 Models Matched on Pass@1 Diverge in Sampling Behavior

Our second central finding is the PSF analogue: first-attempt accuracy does not determine how accuracy scales with more attempts.

PSF fits the data well. Across 41 model-benchmark pairs, Equation 1 achieves a mean R^2 of 0.96. Restricting to the 22 non-degenerate fits (Section 4.2) the mean R^2 is 0.96 and *every* fit exceeds 0.90 (Figures 5, 6, 7, and 8 in Appendix D), confirming that the three-parameter form is a faithful summary of Pass@ k behavior and that the high aggregate R^2 is not an artifact of flat or degenerate curves (full fits in Appendix D, Tables 10 and 11).

Matched Pass@1, divergent Pass@4. Among model pairs matched on Pass@1 (within 0.02), the clearest example is GPT-OSS-120B and GPT-5-mini on mathematics: their Pass@1 differs by 0.006, but their Pass@4 differs by 0.105—more than ten points—and their approach-rate γ differs by a factor of more than twenty. GPT-OSS-120B saturates almost immediately at a lower ceiling, while GPT-5-mini continues to convert additional samples into

additional correct answers. A practitioner choosing between them on Pass@1 alone would see no difference; under repeated sampling, one is substantially more useful than the other. Large Pass@4 gaps at matched Pass@1 are concentrated rather than ubiquitous (Figure 9 in Appendix D)—one pair exceeds a ten-point gap in the strict matched set—but the existence of such pairs demonstrates that Pass@1 and sampling trajectory are decoupled.

Model	α	β	γ	R^2
GPT-OSS-120B	0.828	0.862	30.0 [†]	1.000
GPT-5-mini	0.833	0.958	1.358	0.956

Table 4: PSF parameters for the matched-Pass@1 pair on AIME 2025 with the largest sampling divergence. The two models differ by $\Delta\alpha = 0.005$ but by $\Delta\beta = 0.096$ in saturation ceiling and by a factor of >20 in approach rate γ , yielding an empirical Pass@4 gap of 0.105. [†]GPT-OSS-120B’s γ hit the optimizer upper bound, indicating near-immediate saturation; the true γ is at least this large.

5.4 The Two Instruments Are Independent

If RPC and PSF measured the same underlying property, reporting both would be redundant. They do not. Joining the two instruments on the model-benchmark pairs for which both are defined, the PSF approach-rate γ is at most weakly correlated with each RPC axis (absolute Pearson $r \leq 0.14$ across axes; Figure 10 in Appendix D), and the correlations are not consistent in sign. Reasoning-path consistency and sampling trajectory are thus distinct dimensions of behavior: a model can be highly consistent yet saturate quickly, or inconsistent yet keep improving with samples. This independence is the formal justification for treating RPC and PSF as complementary instruments rather than alternative views of one signal.

5.5 Behavioral Shape Is Domain-Conditional

Finally, a model’s behavioral shape is not fixed—it varies across reasoning domains, and the two RPC axes localize *where* it varies. Figure 2 in Appendix C shows that semantic consistency (RPC-SEM) shifts by benchmark for the same model: a model can score high on law and lower on math or GPQA. By contrast, Figure 3 in Appendix C shows

that structural consistency (RPC-S) varies more sharply across domains-mean within-model range 0.10 (max 0.25 for Gemini-3)-whereas RPC-SEM is comparatively stable (mean within-model range 0.056; max 0.091). Table 5 further shows that designated reasoning models pool slightly higher mean RPC-SEM and RPC-S than non-reasoning models on strict rows, though this comparison is descriptive only. In other words, models tend to preserve the *meaning*-level consistency of their reasoning across domains while varying the *structural* organization of that reasoning by domain. Lexical consistency profiles and mean RPC-SEM by provider family on the law benchmark appear in Appendix C (Figure 4; Table 9). This contrast both reinforces that the axes measure distinct properties and gives a concrete account of how reasoning behavior adapts to task type.

Mode	Mean RPC-SEM	Mean RPC-S	Mean RPC-L	n
Non-reasoning	0.830	0.544	165.0	3,149
Reasoning	0.851	0.594	254.9	849

Table 5: Pooled RPC by reasoning mode on strict rows. “Reasoning” denotes o4-mini and deepseek_r1 (the only reasoning slugs with strict coverage in this merge). Descriptive only; RPC-L is an overlap proxy, not on $[0, 1]$.

5.6 Null Result: Consistency Does Not Predict Cost Stability

We also tested whether reasoning-path consistency predicts the stability of per-attempt cost, hypothesizing that more consistent reasoning would yield more predictable compute. Table 6 reports pooled correlations between per-question cost variance (across four attempts) and each RPC axis over $n = 3,998$ strict rows. We find no meaningful relationship: the correlation between RPC-SEM and cost variance is small (Pearson $r = 0.098$; bootstrap 95% CI $[0.077, 0.129]$) and in the opposite direction to the hypothesis. We report this as a null result; reasoning consistency and cost predictability appear to be unrelated in our data (strict-row pool defined in Appendix C).

Comparison	Pearson r	Spearman ρ
Cost var. vs RPC-SEM	0.098	0.113
Cost var. vs RPC-S	0.011	-0.136
Cost var. vs RPC-L	0.045	0.034

Table 6: Correlation between per-question cost variance (across four attempts) and RPC metrics ($n = 3,998$ strict rows). Bootstrap 95% CI for Pearson r (RPC-SEM vs cost var.): $[0.077, 0.129]$ (500 resamples).

6 Discussion

Capability has a shape. The recurring pattern across our results is that two models can be indistinguishable on Pass@1 and yet behave differently in ways that matter. They can reason along more or less consistent paths (RPC) (Chen et al., 2024; Wan et al., 2025), and they can convert additional attempts into additional correct answers at very different rates (PSF) (Aggarwal et al., 2023). Neither property is visible in first-attempt accuracy, and-because the two instruments are statistically independent-neither is reducible to the other. A single accuracy number projects a multidimensional behavior onto one axis; RPC and PSF recover two of the dimensions it discards (Liu et al., 2024).

Why the matched-Pass@1 framing matters.

We deliberately anchor both central findings on models with near-identical Pass@1. This is the strongest form of the argument: when accuracy is held fixed, any remaining behavioral difference cannot be attributed to one model simply being “better.” (Qi et al., 2023; Yang et al., 2024; Moore et al., 2024) The Claude-Gemini pair reasons with very different consistency at the same accuracy; the GPT-OSS-GPT-5-mini pair scales very differently at the same accuracy. These are not differences in capability magnitude but in capability *shape*, and they are exactly what a scalar metric is constructed to hide.

Practical implications. The two instruments speak to different deployment decisions. PSF is directly actionable for inference-time sampling: a model with a low approach-rate γ and a high ceiling β rewards repeated sampling (Gui et al., 2024; Sun et al., 2024), while a model that saturates immediately does not, and two models tied on Pass@1 can sit on opposite sides of this divide (Brown et al., 2025). Choosing a model for a best-of- k pipeline on Pass@1 alone can therefore be the wrong choice. RPC speaks to reliability and interpretability: a model whose reasoning paths are consistent across attempts is more predictable to audit and debug, whereas low consistency at high accuracy suggests the model reaches correct answers by variable routes-relevant where the reasoning, not just the answer, is subject to review (Pinhanez et al., 2025; Zeng et al., 2026; Zollicoffer et al., 2026; Weng et al., 2023; Hong et al., 2024).

Semantic stability, structural variation. The domain analysis offers a more specific behavioral

observation: across reasoning domains, models tend to hold their *semantic* reasoning consistency roughly fixed while varying the *structural* organization of their reasoning. A plausible reading is that models adapt the scaffolding of a solution to the demands of a domain—more tool calls here, more enumerated steps there—while keeping the underlying conceptual approach comparatively stable (Wen et al., 2025; Nair and Wang, 2024; Rombaut, 2026). That semantic and structural consistency dissociate in this way is further evidence that the RPC axes are not interchangeable.

What the instruments do not claim. RPC and PSF measure properties of behavior, not its quality (Huang et al., 2024; Cruz et al., 2024). High consistency is not inherently good—a model can be consistently wrong—and a high approach-rate is not inherently good, since it may reflect a low ceiling reached quickly (Patwardhan et al., 2025). The instruments are descriptive: they expose structure that accuracy hides, and leave the normative question of which shape is preferable to the deploying practitioner, whose answer will depend on whether they value single-shot accuracy, best-of- k performance, auditability, or cost. The contribution is that this structure is now measurable and comparable across models, rather than collapsed into a number.

Toward shape-aware evaluation. Taken together, the results argue for reporting model behavior as more than a point on a leaderboard. A model card that listed, alongside accuracy, a model’s sampling fingerprint (α, β, γ) and its reasoning-consistency profile would convey decisions-relevant information that accuracy alone omits (Mitchell et al., 2019; Zhang et al., 2025). We do not claim RPC and PSF are the only such dimensions, or the most important ones; we claim that behavioral shape is real, measurable, independent of accuracy, and worth reporting.

7 Conclusion

We argued that Pass@1 accuracy is a lossy summary of a reasoning model’s behavior, and we introduced two instruments that recover structure it discards. Reasoning Path Consistency (RPC) measures how similar a model’s reasoning paths are across repeated attempts, along lexical, structural, and semantic axes, with its semantic axis validated against human judgments of reasoning similarity.

Parametric Sampling Fingerprints (PSF) summarize how a model’s accuracy scales with additional attempts through a compact three-parameter curve that fits empirical Pass@ k behavior with high fidelity.

Across sixteen models and four reasoning benchmarks, both instruments revealed large behavioral differences between models that Pass@1 reports as equivalent: models tied on first-attempt accuracy diverge by up to 0.17 in semantic reasoning consistency and by up to ten points in Pass@4. The two instruments are statistically independent, capturing distinct dimensions of behavior, and both dimensions are domain-conditional—most visibly, models preserve semantic consistency across domains while varying their structural organization. Capability, in short, is not a scalar: it has a shape that accuracy alone cannot express.

Our analysis is bounded—RPC requires accessible reasoning traces, which restricts it to models that expose chain-of-thought, and the strongest matched-accuracy divergences are demonstrated on a subset of benchmarks. But the central claim does not depend on scale: even a single pair of models tied on accuracy yet divergent in reasoning consistency or sampling trajectory is sufficient to show that the scalar hides structure. We see two natural directions: expanding human validation and the matched-accuracy analysis to more benchmarks and models, and developing shape-aware evaluation reporting in which a model’s sampling fingerprint and consistency profile are reported alongside its accuracy. We release our metrics, fitted parameters, and code to support both. Appendices B, C, and D document the experimental roster, RPC computation, and PSF fitting in full.

Limitations

RPC requires reasoning traces. RPC is undefined for models that do not expose chain-of-thought, which excludes the OpenAI-family models from all RPC analyses. Our RPC findings therefore characterize the subset of models that emit reasoning traces, and may not transfer to models whose reasoning is hidden.

Scope of the matched-accuracy analysis. The strongest matched-Pass@1 divergences are demonstrated on a subset of benchmarks—after excluding models without traces, the qualifying matched pairs concentrate on the law benchmark—and the largest PSF Pass@4 gaps are concentrated in a few pairs

rather than pervasive. The findings establish that Pass@1 can decouple from reasoning consistency and from sampling trajectory, not that it does so uniformly across all models and tasks.

Human validation scale. RPC-Sem is validated against 50 human “same-approach” judgments; the lexical and structural axes are validated only on a smaller original set and are weaker proxies for human judgment at this scale. A larger and more balanced human-validation study would be needed to characterize all three axes and to estimate agreement more precisely (Ando and Zhang, 2005).

Four attempts. Both instruments are computed from $k = 4$ attempts per problem. RPC consistency is averaged over six trace pairs, and PSF curves are fit to four Pass@ k points; more attempts would yield more stable consistency estimates and better-constrained curve fits, and might reveal scaling behavior beyond $k = 4$ that a four-point fit cannot capture.

Metric design choices. Each RPC axis embeds specific choices- n -gram order for the lexical axis, the step-type abstraction for the structural axis, and the sentence-embedding model for the semantic axis (Appendix C)- and the PSF form assumes a particular three-parameter functional shape (Appendix D). Alternative choices could yield somewhat different absolute values, though the qualitative finding-that models matched on accuracy diverge under these measures-does not depend on the specific parameterization.

Potential risks. Our instruments (RPC and PSF) are descriptive measures of model behavior and do not generate content or take actions, so the direct risks are limited. The principal risk is misinterpretation: RPC and PSF measure the *shape* of model behavior, not its quality, and treating a high-consistency or fast-saturating fingerprint as evidence of a “better” model would be a misuse-a model can be consistently wrong or saturate quickly at a low ceiling (Section 6). If shape-aware metrics were adopted in model reporting, they should complement rather than replace accuracy and task-grounded evaluation. All evaluation uses public benchmarks and model outputs and involves no personal or sensitive data; our human-validation study collected only similarity judgments over reasoning traces.

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A Code and Data

All four benchmarks (AGIEval LSAT-AR, AIME 2025, GPQA Diamond, MMMU) are publicly available under their respective research licenses; the embedding model all-MiniLM-L6-v2 is released under Apache-2.0. We release our RPC and PSF implementations, fitted parameters, and analysis code at <https://anonymous.4open.science/r/Submission-Code-43D9/>, and the full set of per-attempt reasoning traces and run outputs at https://huggingface.co/datasets/reasoning-fingerprints-anon/REASONING_FINGERPRINTS/tree/main. AI assistants were used for coding assistance and manuscript editing/proofreading. All research design, analysis, and claims are the authors’.

B Models and Benchmarks

B.1 Model Roster

We evaluate sixteen models spanning seven developers. PSF, which requires only per-attempt correctness, is computed over all sixteen. RPC, which requires access to reasoning traces, is computed over the ten models that expose chain-of-thought; the six OpenAI-family models are excluded from RPC for the reason given in Section 3. Table 7 lists the full roster and indicates each model’s inclusion in each instrument.

B.2 Benchmarks

We use four reasoning benchmarks chosen to span distinct domains:

- **Law:** AGIEval LSAT analytical reasoning (agieval_lsar) (Zhong et al., 2023).
- **Mathematics:** AIME 2025 (Mathematical Association of America, 2025).
- **Science:** GPQA Diamond (Rein et al., 2023).
- **Multimodal:** MMMU subsets (Yue et al., 2024) (accounting, clinical_medicine, finance).

Each problem is attempted four times per model under independent sampling.

All four attempts per problem use each provider’s default decoding temperature (we do

Model	Developer	PSF	RPC
Claude 3.7 Sonnet	Anthropic	✓	✓
Claude 4 Opus	Anthropic	✓	✓
Claude 4.5 Sonnet	Anthropic	✓	✓
Gemini 2.5 Pro	Google	✓	✓
Gemini 3 Preview	Google	✓	✓
DeepSeek-R1	DeepSeek	✓	✓
Grok-3-mini	xAI	✓	✓
MiniMax M2	MiniMax	✓	✓
Qwen3-235B	Qwen	✓	✓
Qwen3-30B	Qwen	✓	✓
GPT-5	OpenAI	✓	-
GPT-5-mini	OpenAI	✓	-
GPT-5-nano	OpenAI	✓	-
GPT-5.1	OpenAI	✓	-
o4-mini	OpenAI	✓	-
GPT-OSS-120B	OpenAI	✓	-

Table 7: Model roster. PSF is computed over all sixteen models; RPC over the ten that expose reasoning traces. OpenAI-family models return only final answers and are excluded from RPC (Section 3); any RPC-like values computed from their final-answer text are not comparable to trace-based RPC and are not reported.

not override the sampling temperature); for configurable chat models this is temperature 1.0, while certain reasoning models do not expose a configurable temperature through their API. We use stochastic (non-greedy) decoding throughout, since both instruments depend on variation across repeated attempts.

B.3 Per-Benchmark Coverage

Not every model has complete coverage on every benchmark, due to trace availability and attempt completeness. Table 8 reports, for each instrument, the number of model-benchmark cells contributing to the analysis.

Benchmark	RPC cells	PSF fits
AGIEval LSAT-AR	2,159	12
AIME 2025	403	16
GPQA Diamond	735	4
MMMU (subsets)	701	9
Total	3,998	41

Table 8: Per-benchmark coverage for each instrument. **RPC cells:** strict model-question rows with RPC-L, RPC-S, and RPC-SEM all defined (from rpc_task2_merged.csv). **PSF fits:** model-benchmark pairs with a fitted PSF curve (from Squad C P1 outputs). MMMU aggregates accounting, clinical medicine, and finance subsets.

C RPC Computation Details

This appendix supplements Section 5.1 and Section 5.5 with RPC metric definitions, domain heatmaps, the strict-row filter, human-validation protocol, and supplementary breakdowns by family

and lexical consistency. Figure 4 uses a separate scale from Figures 2 and 3 (below) because RPC-L is not normalized to $[0, 1]$.

Human validation. For Experiment R1, we sampled 50 trace pairs and collected human judgments of whether the two paths take the same approach. Annotators were shown two reasoning traces for the same problem and asked: Do these two solutions take the same overall approach? Answer yes or no. No additional guidance or risk disclaimer was provided, as annotators judged only model-generated traces over public benchmarks. Labels were aggregated to a mean score per pair (human_same_approach_mean) and correlated with each RPC axis; full results appear in Table 1. Only RPC-SEM aligns significantly with human judgment at this scale (Spearman $\rho = 0.34$, $p = 0.017$). The judgments were provided by student researchers affiliated with the project as part of their research involvement; no crowdsourcing platform was used and annotators were not separately compensated per annotation. Annotators were members of the research team and were aware that their judgments would be used for this research.

RPC-L. Mean overlap of token 3-grams between each pair of the four reasoning traces, averaged over the six pairs (Stolcke et al., 2000; Gusfield, 1997). Reported as a raw overlap count (not normalized to $[0, 1]$), so RPC-L magnitudes are comparable across traces of similar length but should not be read as a probability. Domain-level lexical profiles are shown in Figure 4; do not compare its color scale directly to Figures 2 or 3.

RPC-S. Similarity of the step-type sequences of the four traces, computed with the same n -gram construction applied to step-type tokens rather than surface tokens (Aho and Ullman, 1972; Rasooli and Tetreault, 2015). Model-benchmark structural profiles are visualized in Figure 3.

RPC-SEM. Mean pairwise cosine similarity of the four trace embeddings, using sentence-transformers/all-MiniLM-L6-v2 (Wang et al., 2020). Embeddings are L2-normalized, so cosine similarity equals the dot product. Domain-level semantic profiles appear in Figure 2.

Strict rows. All metric-metric correlations (Table 2) and pooled analyses in Section 5.6 are computed on the *strict* set of $n = 3,998$ model-question

cells for which all three RPC axes are defined (has_rpc_s = 1, has_rpc_l = 1, and valid RPC-SEM). Human-validation correlations (Table 1) use the 50 labeled pairs joined to this merge on source_path.

Family profiles on AGIEval LSAT-AR. Table 9 reports mean RPC-SEM by provider family on the law benchmark, complementing the reasoning-mode breakdown in Table 5. Row counts are model-question cells, not number of models; OpenAI-family RPC is computed only on questions with available CoT text (see limitations).

Family	Mean RPC-SEM	n (cells)
DeepSeek	0.920	230
Anthropic	0.905	690
MiniMax	0.822	228
OpenAI	0.796	749
Google	0.759	262

Table 9: Mean RPC-SEM by model family on AGIEval LSAT-AR. Row counts are model-question cells, not number of models. OpenAI-family RPC is computed on questions with available CoT text (see limitations).

D PSF Fitting Details

We fit Equation 1 by least squares per model-benchmark pair, with $\alpha, \beta \in [0, 1]$ and $\gamma > 0$. We fit each curve with non-linear least squares (SciPy curve_fit, trust-region reflective), with α, β bounded to $[0, 1]$ and $\gamma > 0$. Of 41 fits, 22 are non-degenerate under the criteria in Section 4.2 ($|\alpha - \beta| > 0.02$, γ not at the optimizer boundary, ≥ 30 problems). Mean R^2 is 0.96 over all fits and 0.96 over the non-degenerate subset, with every non-degenerate fit exceeding 0.90. Degenerate fits arise predominantly when accuracy is near-constant across k , so that the curve is underdetermined; we exclude these from interpretive claims. Figures 5, 6, 7, and 8 plot empirical and fitted Pass@ k curves by benchmark; Figure 9 summarizes matched-Pass@1 Pass@4 gaps; and Figure 10 visualizes independence between PSF approach rate and RPC-SEM.

Full fitted parameters appear in Tables 10 and 11.

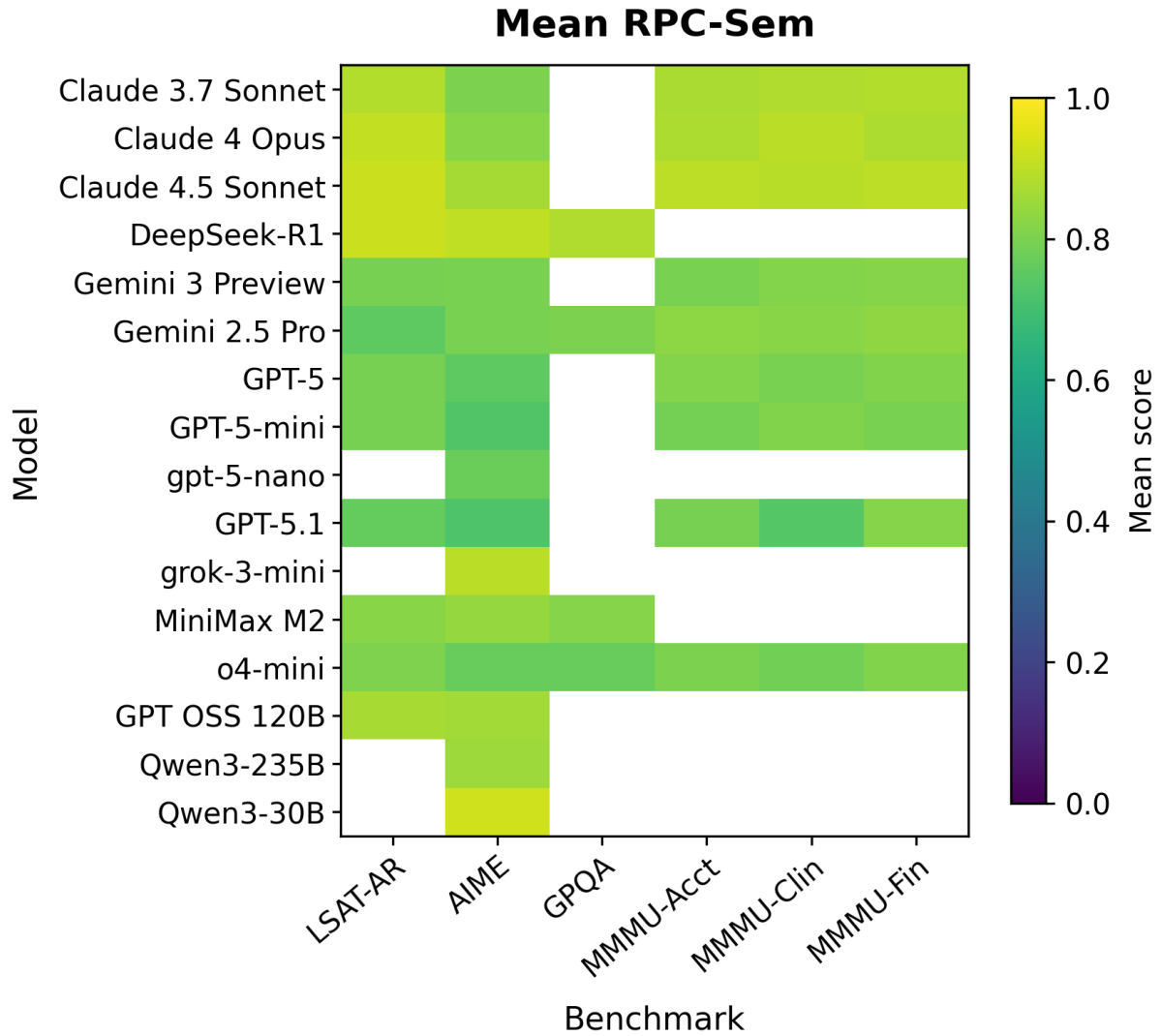


Figure 2: Domain-conditioned semantic path consistency (RPC-SEM) by model and benchmark. Each cell is mean RPC-SEM over strict model-question rows; blank cells indicate no strict coverage. The same model’s consistency shifts across domains (e.g., law vs. math vs. GPQA vs. MMMU). Across models with coverage on ≥ 2 benchmarks, mean within-model RPC-SEM range is 0.056 (max 0.091).

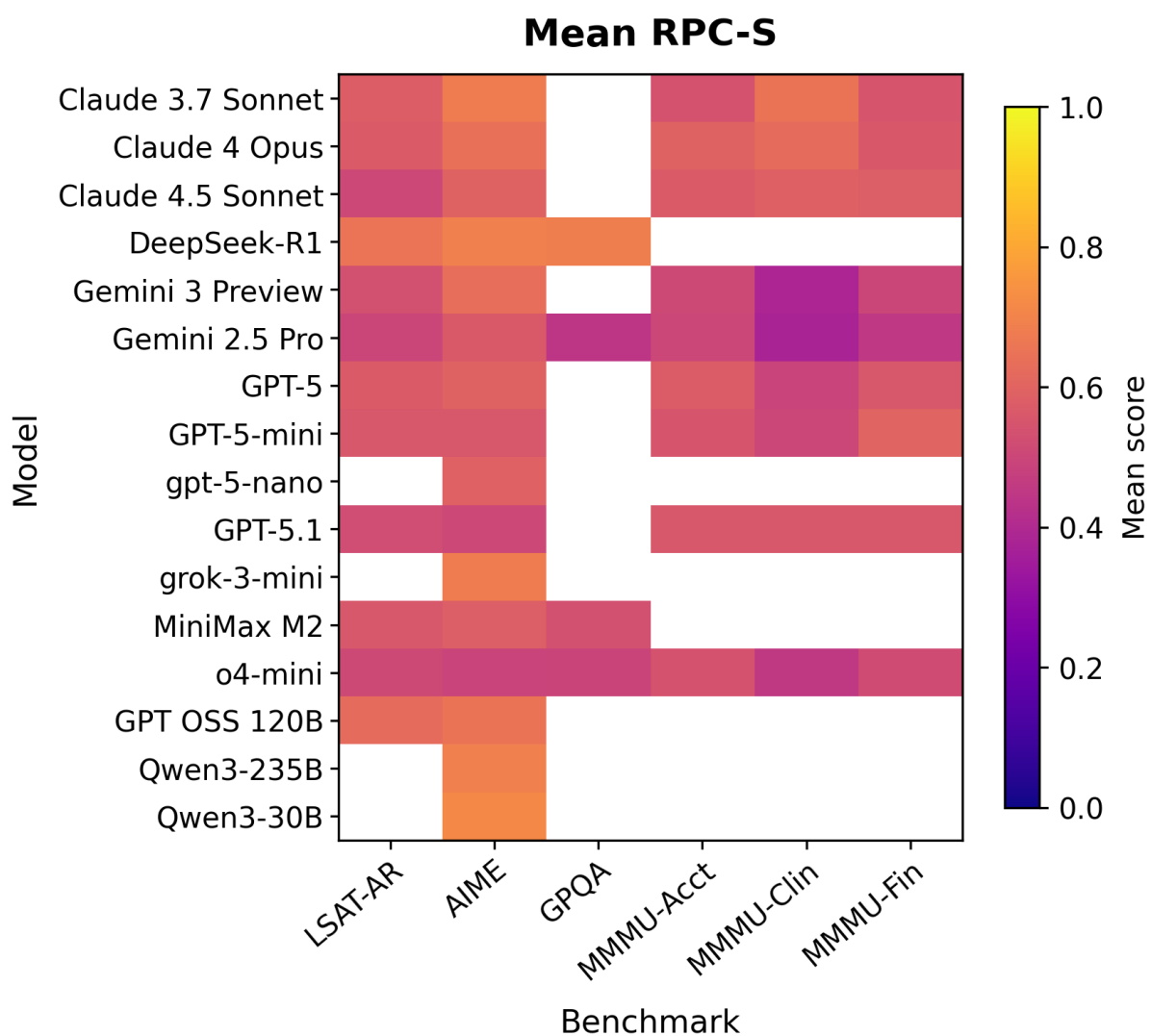


Figure 3: Structural path consistency (RPC-S) by model and benchmark. Structural consistency varies more across domains than semantic consistency: mean within-model range 0.10 (max 0.25 for Gemini-3).

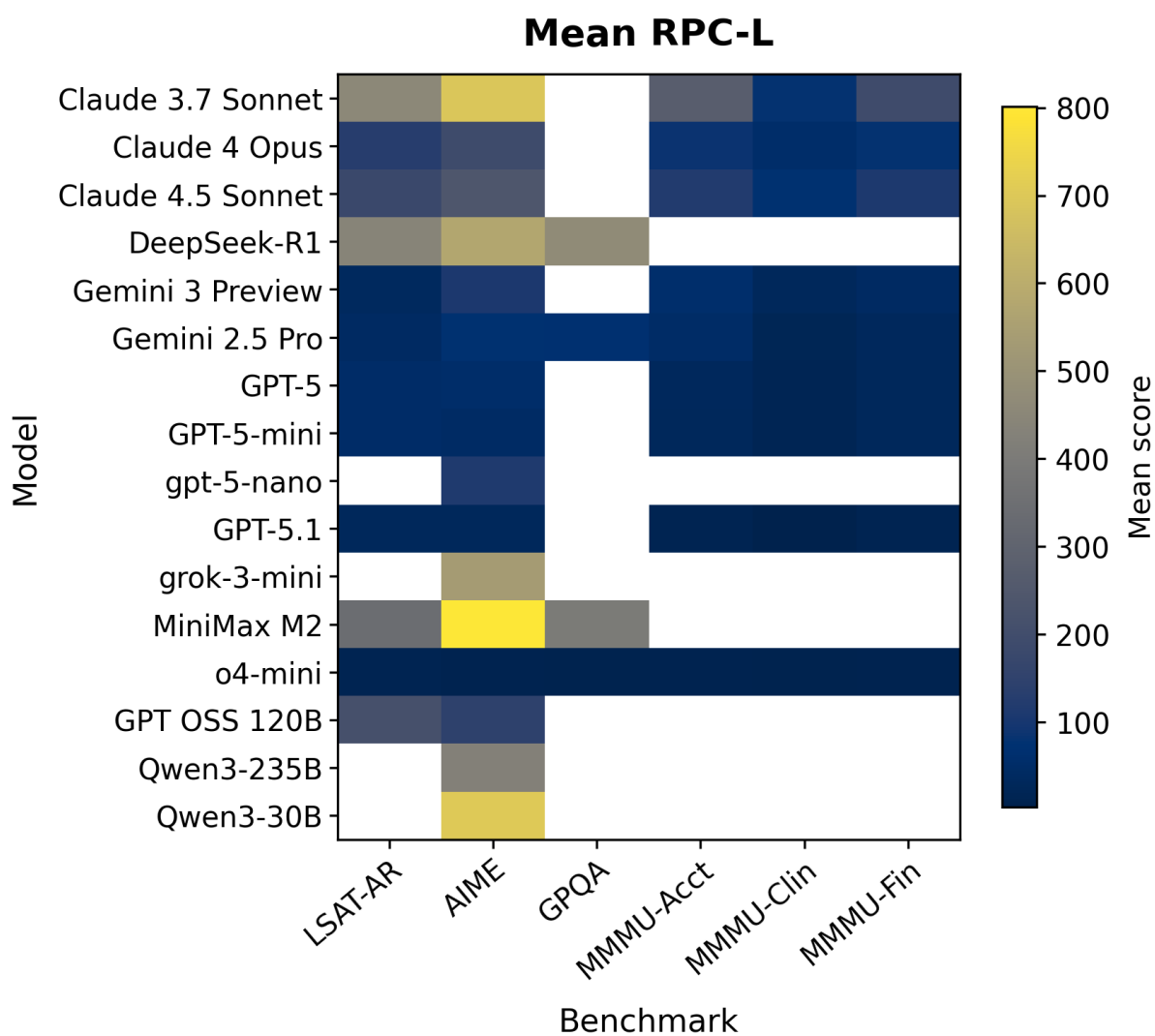


Figure 4: Lexical path consistency (RPC-L) by model and benchmark. RPC-L is mean 6-pair CoT n -gram overlap (not normalized to $[0, 1]$); use a separate scale from RPC-SEM and RPC-S.

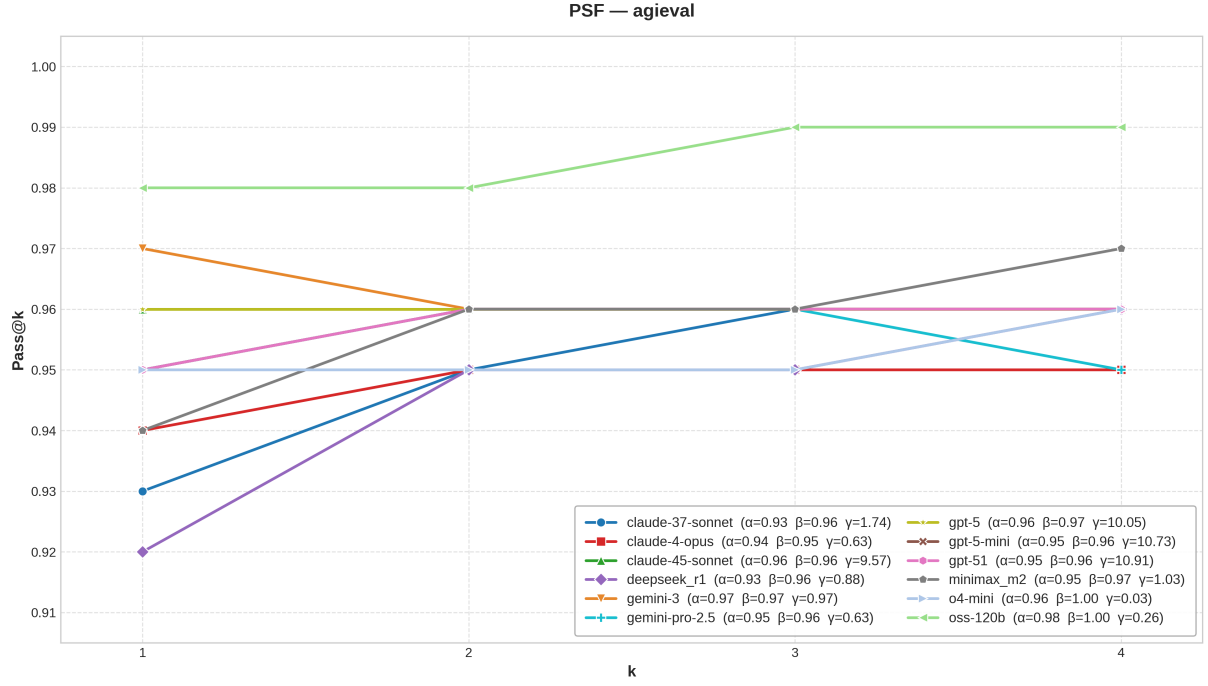


Figure 5: PSF curve fits on AGIEval LSAT-AR. Empirical Pass@ k for $k = 1 \dots 4$ (points) and the fitted three-parameter curve (Equation 1) for each model on this benchmark.

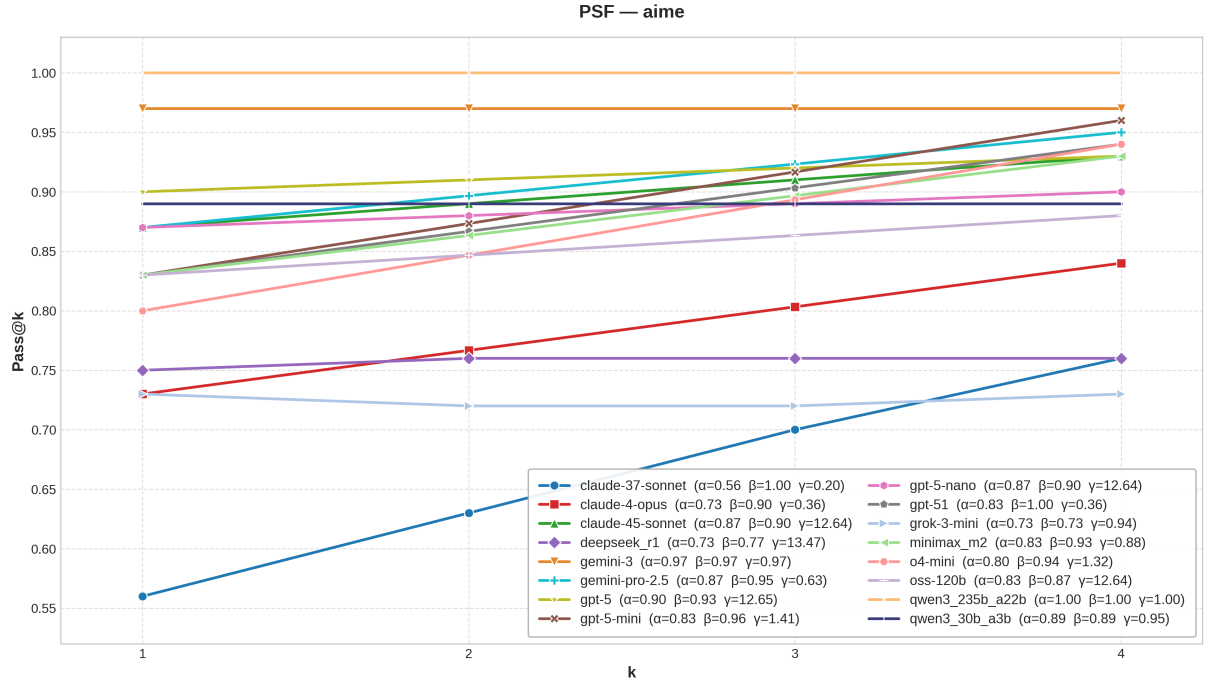


Figure 6: PSF curve fits on AIME 2025 (same notation as Figure 5).

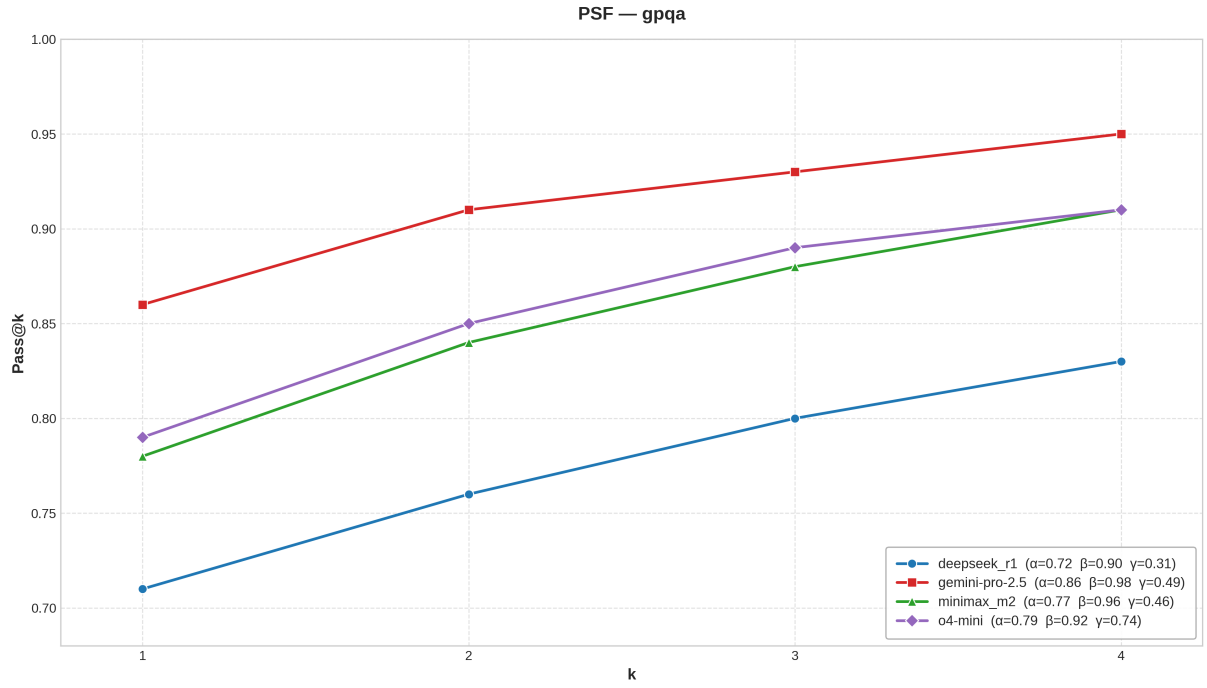


Figure 7: PSF curve fits on GPQA Diamond (same notation as Figure 5).

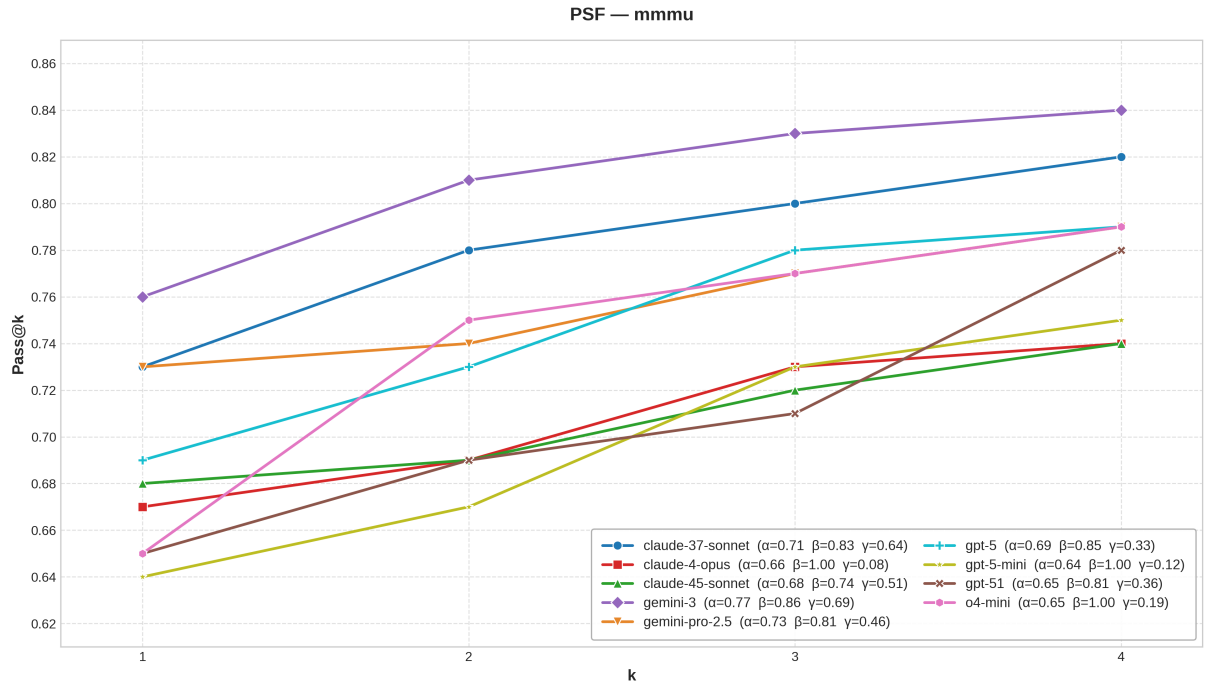


Figure 8: PSF curve fits on MMMU subsets (same notation as Figure 5). Mean $R^2 = 0.96$ over all 41 model-benchmark fits in this appendix.

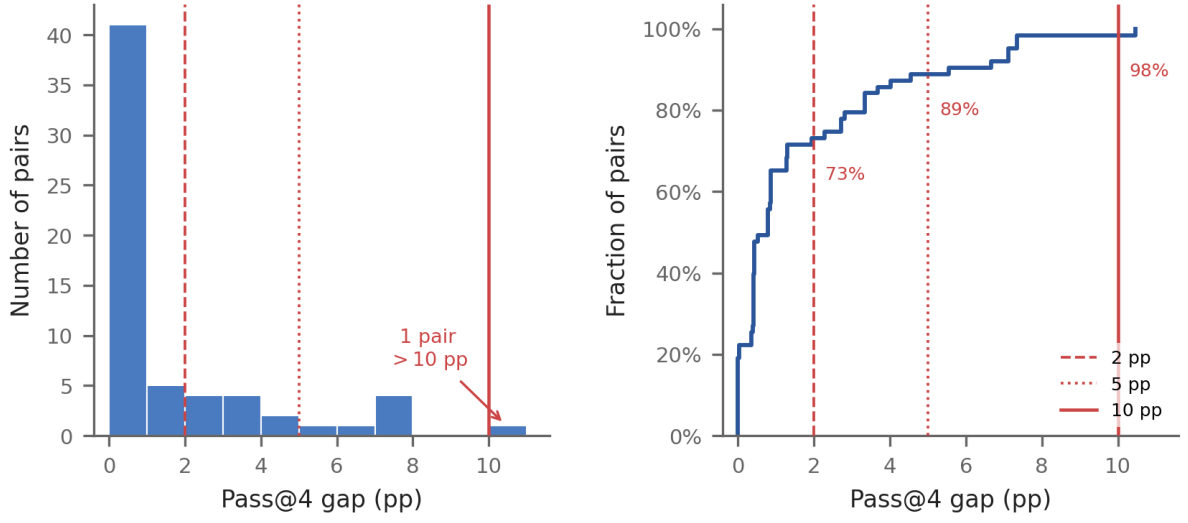


Figure 9: Distribution of Pass@4 gaps across all 63 qualifying matched-Pass@1 pairs ($|\Delta\text{Pass@1}| \leq 0.02$). *Left:* histogram of gap magnitudes in percentage points; *right:* empirical CDF. Dashed, dotted, and solid red lines mark the 2, 5, and 10 pp thresholds respectively. Large gaps are rare: 27.0% of pairs exceed 2 pp, 11.1% exceed 5 pp, and only 1.6% (a single pair, GPT-OSS-120B vs. GPT-5-mini on AIME) exceed 10 pp. The median gap is 0.79 pp, confirming that large Pass@4 divergence at matched Pass@1 is concentrated rather than pervasive.

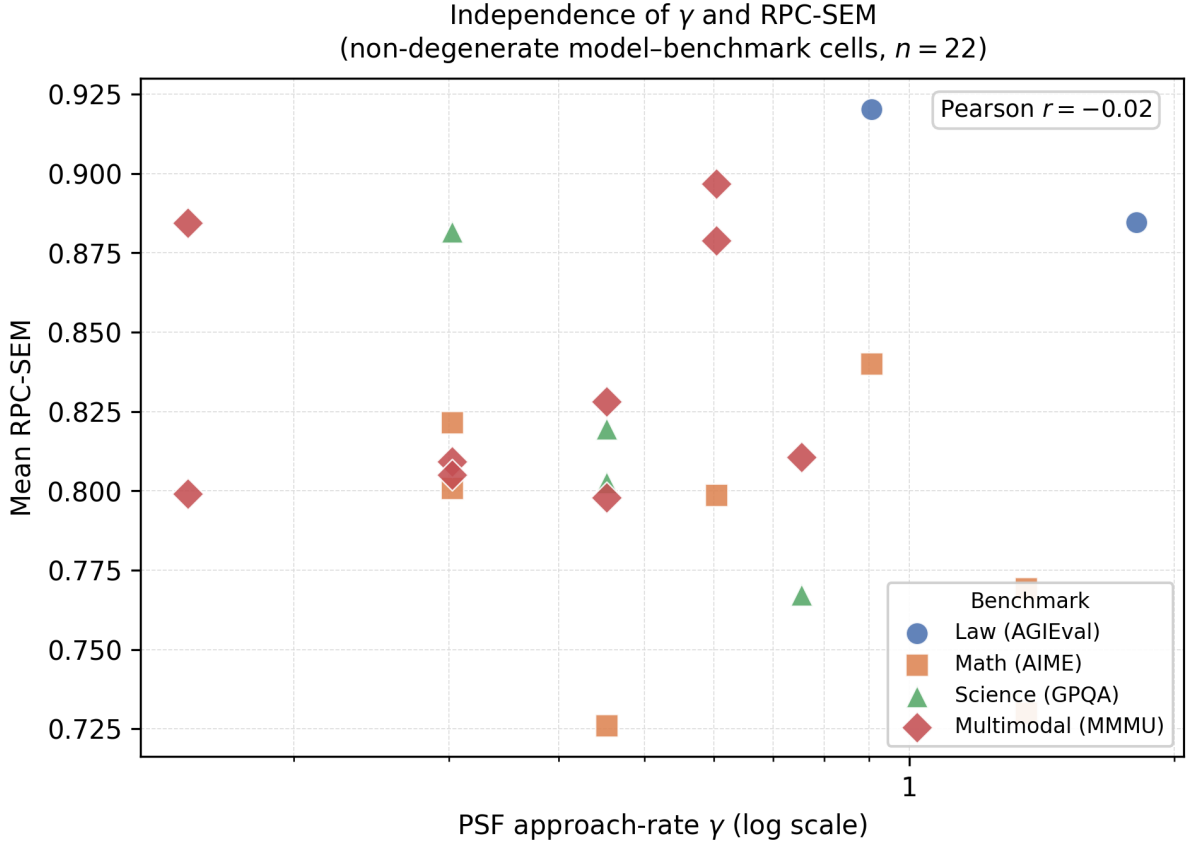


Figure 10: Independence of PSF approach-rate γ and semantic reasoning consistency RPC-SEM. Each point is one model-benchmark cell (non-degenerate fits only; $n = 22$ cells); the x -axis is log-scaled because γ spans several orders of magnitude. The near-zero Pearson r (annotated on plot; absolute value ≤ 0.14 across all RPC axes) confirms that reasoning-path consistency and sampling trajectory are statistically independent dimensions of model behavior: a model can be highly consistent yet saturate quickly, or inconsistent yet keep improving with additional samples.

Model	Bench.	n	α	β	γ	R^2	ND
Claude 3.7 Sonnet	AGIEval	230	0.930	0.957	1.810	0.999	✓
Claude 3.7 Sonnet	AIME	30	0.567	0.880	0.302	0.978	✓
Claude 3.7 Sonnet	MMMU	90	0.711	0.838	0.604	0.964	✓
Claude 4 Opus	AGIEval	230	0.943	0.954	0.604	0.969	
Claude 4 Opus	AIME	30	0.733	0.914	0.302	0.923	✓
Claude 4 Opus	MMMU	90	0.667	0.849	0.152	0.915	✓
Claude 4.5 Sonnet	AGIEval	230	0.957	0.961	29.849	1.000	
Claude 4.5 Sonnet	AIME	30	0.867	0.900	30.0 [†]	1.000	
Claude 4.5 Sonnet	MMMU	90	0.678	0.739	0.604	0.910	✓
DeepSeek-R1	AGIEval	230	0.926	0.964	0.905	0.929	✓
DeepSeek-R1	AIME	30	0.733	0.767	30.0 [†]	1.000	
DeepSeek-R1	GPQA	198	0.717	0.907	0.302	0.994	✓
Gemini 3	AGIEval	32	0.969	0.969	0.001 [†]	1.000	
Gemini 3	AIME	30	0.967	0.967	0.001 [†]	1.000	
Gemini 3	MMMU	90	0.767	0.853	0.755	0.999	✓
Gemini 2.5 Pro	AGIEval	230	0.952	0.963	0.604	0.969	
Gemini 2.5 Pro	AIME	30	0.867	0.951	0.604	0.969	✓
Gemini 2.5 Pro	GPQA	198	0.859	0.991	0.453	0.976	✓
Gemini 2.5 Pro	MMMU	90	0.733	0.813	0.453	0.944	✓

Table 10: Full PSF parameter table, rows 1-19 (continued in Table 11). α : Pass@1; β : saturation ceiling; γ : approach rate. ND (✓) = non-degenerate fit ($|\alpha - \beta| > 0.02$, γ not at optimizer boundary, $n \geq 30$). [†] γ at optimizer boundary (≈ 0 or ≈ 30).

Model	Bench.	n	α	β	γ	R^2	ND
GPT-5	AGIEval	230	0.957	0.965	30.0 [†]	1.000	
GPT-5	AIME	30	0.900	0.933	30.0 [†]	1.000	
GPT-5	MMMU	90	0.689	0.858	0.302	1.000	✓
GPT-5-mini	AGIEval	230	0.952	0.961	30.0 [†]	1.000	
GPT-5-mini	AIME	30	0.833	0.958	1.358	0.956	✓
GPT-5-mini	MMMU	90	0.644	0.939	0.152	0.911	✓
GPT-5-nano	AIME	29	0.862	0.897	30.0 [†]	1.000	
GPT-5.1	AGIEval	228	0.947	0.965	30.0 [†]	1.000	
GPT-5.1	AIME	30	0.833	0.974	0.453	0.920	✓
GPT-5.1	MMMU	88	0.648	0.794	0.453	0.986	✓
Grok-3-mini	AIME	8	1.000	1.000	0.001 [†]	1.000	
MiniMax M2	AGIEval	228	0.952	0.966	1.056	0.992	
MiniMax M2	AIME	30	0.833	0.931	0.905	0.929	✓
MiniMax M2	GPQA	198	0.768	0.966	0.453	0.999	✓
o4-mini	AGIEval	230	0.957	0.964	0.152	0.470	
o4-mini	AIME	30	0.800	0.938	1.358	0.997	✓
o4-mini	GPQA	195	0.790	0.917	0.755	0.990	✓
o4-mini	MMMU	88	0.659	0.891	0.302	0.974	✓
GPT-OSS-120B	AGIEval	84	0.976	0.996	0.302	0.751	
GPT-OSS-120B	AIME	29	0.828	0.862	30.0 [†]	1.000	
Qwen3-235B	AIME	6	1.000	1.000	0.001 [†]	1.000	
Qwen3-30B	AIME	6	1.000	1.000	0.001 [†]	1.000	

Table 11: Full PSF parameter table, rows 20-41 (continued from Table 10). See Table 10 caption for column definitions. Of 41 total fits, 22 are non-degenerate (ND = ✓); mean $R^2 = 0.96$ over both subsets.